

TO Identify Built-ins, Create Dispatch Table, Execute In- Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

Code:

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>

void pwd_command() {
    char path[100];

    getcwd(path, sizeof(path));
    printf("Current directory: %s\n", path);
}

void cd_command(char *path) {
    if (chdir(path) == 0)
        printf("Directory changed successfully\n");
    else
        printf("Invalid directory\n");
}

void exit_command() {
    printf("Exiting shell...\n");
    exit(0);
}

/* Dispatch table */
struct Command {
    char name[20];
    int type;
};

struct Command commands[] = {
    {"pwd" 1}
```

```

        {"cd", 2},
        {"exit", 3}
    };

int main() {
    char input[100];
    char command[20];
    char argument[80];
    int i, found;

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        /* Remove newline */
        input[strcspn(input, "\n")] = '\0';

        /* Separate command and argument */
        sscanf(input, "%s %s", command, argument);

        found = 0;

        /* Search dispatch table */
        for (i = 0; i < 3; i++) {

            if (strcmp(command, commands[i].name) == 0) {

                found = 1;

                if (commands[i].type == 1) {
                    pwd_command();
                }
            }
        }
    }
}

```

```

/* Search dispatch table */
for (i = 0; i < 3; i++) {

    if (strcmp(command, commands[i].name) == 0) {

        found = 1;

        if (commands[i].type == 1) {
            pwd_command();
        }

        else if (commands[i].type == 2) {
            cd_command(argument);
        }

        else if (commands[i].type == 3) {
            exit_command();
        }

        break;
    }
}

if (found == 0) {
    printf("Invalid command\n");
}

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 4$ ./task4
myshell> pwd
Current directory: /home/geya/OS/Skill/Week 4
myshell> cd ..
Directory changed successfully
myshell> pwd
Current directory: /home/geya/OS/Skill
myshell> exit
Exiting shell...
geya@LenovoL480:~/OS/Skill/Week 4$ |

```